Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT PROGRESS 3

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Title: Pancreatic Cancer Prediction

Group Name	Group 10
Group Members	1. RAVINESH A/L MARAN (A23CS0175) 2. WELSON WOONG LU BIN (A23CS0196) 3. BERNICE LOU MIN YUN (A23CS0056)
Section	01
Lecturer's Name	DR. SEAH CHOON SEN

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1.0 Exploratory Data Analysis(Pandas/Numpy)

Exploratory Data Analysis (EDA) is an important phase in the process of realizing the trends, associations, and distributions within a dataset, prior to the implementation of machine learning models. There are Python libraries used to summarize the data including Pandas, NumPy, Matplotlib, and Seaborn, which can be used to visualize the distributions, detect anomalies, and identify relationships between variables.

The EDA objectives of this project are to determine the relationship between demographic variables, clinical variables, and lifestyle variables and Survival_Status and Survival_Time_Months among pancreatic cancer patients.

1.1 Descriptive Statistics

his code performs descriptive statistical analysis on both numerical and categorical variables to understand the overall characteristics of the dataset. The describe( ) function is applied to the age and survival_time_months variables to generate summary statistics, including count, mean, standard deviation, and quartiles. The results indicate that these variables have been normalized, as shown by mean values close to zero and standard deviations close to one.

For categorical and binary variables, a list of relevant columns is defined and their frequency distributions are calculated using the value_counts( ) function. This allows observation of how data is distributed across categories such as gender, cancer stage at diagnosis, treatment type, smoking history, and survival status.

```python
# descriptive statistic

desc_numeric = df[['age', 'survival_time_months']].describe()
print(desc_numeric)

#binary variable
categorical_cols = [
    'gender',
    'stage_at_diagnosis',
    'treatment_type',
    'smoking_history',
    'survival_status'
]

for col in categorical_cols:
    print(f"\n{col} distribution:")
    print(df[col].value_counts())
```

The output shows that male patients slightly outnumber female patients, and a larger proportion of patients are diagnosed at later stages, particularly Stage IV. The survival status distribution also reveals an imbalance between surviving and non-surviving patients. Overall, this analysis provides an initial understanding of data distribution and potential class imbalance, which is important for subsequent exploratory analysis and model development.

```
                 age  survival_time_months
count  4.999600e+04          4.999600e+04
mean   3.612688e-16         -4.093054e-17
std    1.000010e+00          1.000010e+00
min   -3.463106e+00         -1.144306e+00
25%   -6.558025e-01         -7.007351e-01
50%    4.602341e-02         -3.458782e-01
75%    6.475885e-01          4.525496e-01
max    2.552545e+00          4.001118e+00

gender distribution:
gender
male      25959
female    24037
Name: count, dtype: int64

stage_at_diagnosis distribution:
stage_at_diagnosis
Stage_IV     19918
Stage_III    14968
Stage_II     10173
Stage_I       4937
Name: count, dtype: int64

treatment_type distribution:
treatment_type
Chemotherapy    24907
Radiation       15129
Surgery          9960
Name: count, dtype: int64
```

```
smoking_history distribution:
smoking_history
0    35019
1    14977
Name: count, dtype: int64

survival_status distribution:
survival_status
0    43574
1     6422
Name: count, dtype: int64
```

1.2 Basic Of Grouping

This code applies basic grouping analysis using the groupby( ) function to examine how average survival time differs across selected patient characteristics. The mean of survival_time_months is calculated for each group, including age group, stage at diagnosis, treatment type, and smoking history. As the survival time values were standardized earlier, the results represent relative differences around the overall mean.

```python
# basic grouping
# Survival (months)=(z-score×σ (std))+ μ(mean)

age_group_survival = df.groupby('age_group', observed=True)['survival_time_months'].mean()
print(age_group_survival)

print("\n")
stage_survival = df.groupby('stage_at_diagnosis')['survival_time_months'].mean()
print(stage_survival)

print("\n")
treatment_survival = df.groupby('treatment_type')['survival_time_months'].mean()
print(treatment_survival)

print("\n")
smoking_survival = df.groupby('smoking_history')['survival_time_months'].mean()
print(smoking_survival)
```

The grouped results show that patients diagnosed at earlier stages (Stage I and Stage II) have higher average survival times, while those diagnosed at later stages (Stage III and Stage IV) exhibit lower average survival times. Differences in survival time across treatment types and smoking history are present but relatively small. Overall, this grouping analysis provides an initial understanding of how survival time varies among different patient groups and supports further statistical analysis in the next stage.

```
age_group
Below-Average    0.001027
Average         -0.003200
Above-Average    0.002046
Name: survival_time_months, dtype: float64


stage_at_diagnosis
Stage_I      1.884412
Stage_II     0.576556
Stage_III   -0.082687
Stage_IV    -0.699417
Name: survival_time_months, dtype: float64


treatment_type
Chemotherapy    0.000430
Radiation       0.007325
Surgery        -0.012202
Name: survival_time_months, dtype: float64


smoking_history
0   -0.002257
1    0.005276
Name: survival_time_months, dtype: float64
```

1.3 ANOVA

This code applies a one-way ANOVA (Analysis of Variance) test to examine whether the mean survival time differs significantly across cancer stages at diagnosis. The f_oneday( ) function from the SciPy library is used for this purpose.

First, the survival time data is grouped based on the unique values of stage_at_diagnosis. For each cancer stage, the corresponding survival_time_months values are extracted and stored in a list. These groups are then passed into the ANOVA function to compare the mean survival times across all stages simultaneously.

```python
#  anova

from scipy.stats import f_oneway

stage_groups = [
    df[df['stage_at_diagnosis'] == stage]['survival_time_months']
    for stage in df['stage_at_diagnosis'].unique()
]

f_stat_stage, p_value_stage = f_oneway(*stage_groups)

print("ANOVA: Survival Time vs Stage at Diagnosis")
print("F-statistic:", f_stat_stage)
print("p-value:", p_value_stage)
```

The output shows a very large F-statistic and a p-value of 0.0, which is far below the standard significance level of 0.05. This indicates that there is a statistically significant difference in mean survival time between at least two cancer stages.

Overall, the ANOVA results confirm that the stage at diagnosis has a significant effect on patient survival time, supporting the earlier grouping analysis and providing statistical evidence for differences observed across cancer stages.

```
ANOVA: Survival Time vs Stage at Diagnosis
F-statistic: 26644.673839144925
p-value: 0.0
```

1.4 Correlation

This code prepares the dataset for correlation analysis by converting all variables into a numerical format and ensuring they are on a comparable scale. Since correlation calculations require numerical inputs, categorical variables are first encoded using LabelEncoder. Each unique category is transformed into a numerical label, allowing these variables to be included in the correlation analysis.

After encoding, numerical columns are identified, and selected features are standardized using StandardScaler. This step ensures that variables with different ranges and units do not disproportionately influence the correlation results. The target variable (survival_status) is excluded from scaling to preserve its original meaning.

```
# correlation analysis

label_encoders = {}
for col in df.select_dtypes(include=['object']).columns:
    label_encoders[col] = LabelEncoder()
    df[col] = label_encoders[col].fit_transform(df[col])

original_objects_cols = list(label_encoders.keys())
num_col = df.select_dtypes(include=['int64','float64']).columns

num_col_scale = [col for col in num_col if col not in original_objects_cols and col != 'Survival_Status']

scaler = StandardScaler()
df[num_col_scale] = scaler.fit_transform(df[num_col_scale])

df[num_col_scale].head()
```
0.1s

The output preview confirms that the selected variables have been successfully transformed into standardized numerical values, with values centered around zero. This processed dataset is now suitable for computing correlation coefficients, enabling analysis of the relationships between clinical, lifestyle, and survival-related variables.

Overall, this preprocessing step ensures accurate and meaningful correlation analysis by maintaining consistency across all features and reducing bias caused by scale differences.

	age	smoking_history	obesity	diabetes	chronic_pancreatitis	family_history	hereditary_condition	jaundice	abdominal_discomfort	back_pain	weight_loss
0	-0.054237	-0.653974	-0.574702	-0.499994	-0.332013	-0.422868	-0.22807	-0.498806	-0.649208	-0.581754	-0.733748
1	1.249154	1.529113	1.740032	-0.499994	-0.332013	-0.422868	-0.22807	-0.498806	-0.649208	-0.581754	-0.733748
2	0.647588	-0.653974	-0.574702	-0.499994	-0.332013	-0.422868	-0.22807	-0.498806	-0.649208	-0.581754	1.362866
3	-0.856324	-0.653974	-0.574702	-0.499994	-0.332013	2.364804	-0.22807	2.004788	-0.649208	-0.581754	-0.733748
4	1.750458	-0.653974	-0.574702	-0.499994	-0.332013	2.364804	-0.22807	-0.498806	-0.649208	-0.581754	-0.733748

hereditary_condition	jaundice	abdominal_discomfort	back_pain	weight_loss	development_of_type2_diabetes	survival_time_months	survival_status	alcohol_consumption
-0.22807	-0.498806	-0.649208	-0.581754	-0.733748	-0.494111	-0.079736	-0.383903	-0.660089
-0.22807	-0.498806	-0.649208	-0.581754	-0.733748	2.023836	-0.079736	-0.383903	1.514947
-0.22807	-0.498806	-0.649208	-0.581754	1.362866	2.023836	-0.966878	2.604825	-0.660089
-0.22807	2.004788	-0.649208	-0.581754	-0.733748	2.023836	-0.700735	-0.383903	1.514947
-0.22807	-0.498806	-0.649208	-0.581754	-0.733748	-0.494111	-0.434592	2.604825	-0.660089

This code performs **correlation analysis** to examine the relationships between variables in the dataset. Since correlation calculations require numerical inputs, categorical variables were first converted into numeric form using LabelEncoder. This transformation assigns a unique numerical label to each category, allowing categorical features to be included in the correlation computation.

After encoding, the **Pearson correlation coefficient** was calculated using the df.corr(method='pearson') function. Pearson correlation measures the strength and direction of linear relationships between pairs of variables, with values ranging from −1 to +1. A value close to +1 indicates a strong positive relationship, while a value close to −1 indicates a strong negative relationship.

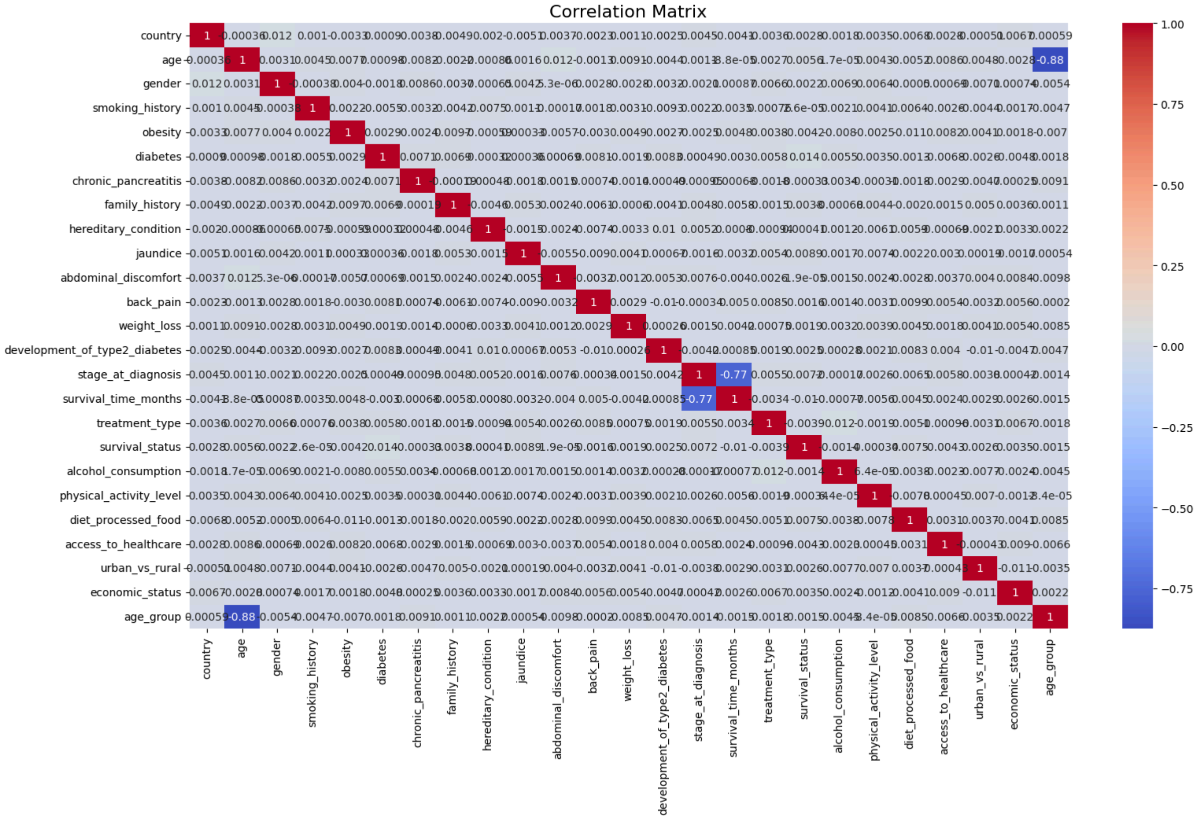
```
# correlation analysis

label_encoders = {}
for column in df.select_dtypes(include=['object', 'category']).columns:
    label_encoders[column] = LabelEncoder()
    df[column] = label_encoders[column].fit_transform(df[column])

correlation_matrix = df.corr(method='pearson')

plt.figure(figsize=(18, 10))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", cbar=True)
plt.title("Correlation Matrix", fontsize=16)
plt.show()
```

✓ 1.1s

The resulting correlation matrix is visualized using a heatmap, where warmer colors represent positive correlations and cooler colors represent negative correlations. From the heatmap, a strong negative correlation is observed between stage at diagnosis and survival time, indicating that patients diagnosed at later stages tend to have shorter survival times. A strong negative correlation is also visible between age and age group, which is expected since age group is derived directly from the age variable.

Most other variables show weak correlations with survival time, suggesting that their relationships are either limited or non-linear. This indicates that survival outcome is influenced by multiple interacting factors rather than a single dominant variable.

Overall, this correlation analysis helps identify important relationships, detect potential multicollinearity, and guide feature selection for subsequent model development.


Correlation Matrix
country
age
gender
smoking_history
obesity
diabetes
chronic_pancreatitis
family_history
hereditary_condition
jaundice
abdominal_discomfort
back_pain
weight_loss
development_of_type2_diabetes
stage_at_diagnosis
survival_time_months
treatment_type
survival_status
alcohol_consumption
physical_activity_level
diet_processed_food
access_to_healthcare
urban_vs_rural
economic_status
age_group
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75